

Lecture 11 Notes

Departure from the Assumptions of the Normal Linear Regression Model

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1 Departure from the Assumptions of Normal Linear Regression Model

The normal linear regression model is written as

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}; \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 I).$$

1.1 Assumptions

- (i) $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \forall i \neq j$
- (ii) $\text{Var}(\varepsilon_i) = \sigma^2 \quad \forall i$
- (iii)

X is non-random.
- (iv) $E(\varepsilon_i) = 0 \quad \forall i$
- (v) ε_i 's are normally distributed / independent.
- (vi) $(X'X)^{-1}$ exists.
- (vii)* linearity.

Note

If $x_{(j)}$'s are centered then $X'X$ is the variance-covariance matrix of X

$$\left(\text{Cov}(x_{(i)}, x_{(j)}) \right).$$

1.2 Questions to Explore

Now we will see:

- What happens if any of these assumptions is violated.
- How to verify the assumptions.
- What to do if one of these assumptions does not hold.

2 Unbiasedness

Assuming $E(\varepsilon_i) = 0 \forall i$.

Suppose we consider the predictors $\{x_1, \dots, x_p\}$. But there are a few other predictors $\{z_1, \dots, z_k\}$ which we fail to consider.

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad E(\boldsymbol{\varepsilon}) = \mathbf{0} \quad (\text{optional / must - error term})$$

2.1 The Actual vs. Working Model

The actual model should be:

$$\mathbf{Y} = \underbrace{X}_{n \times (p+1)} \boldsymbol{\beta} + \underbrace{Z}_{n \times k} \boldsymbol{\gamma} + \boldsymbol{\varepsilon}^*, \quad E(\boldsymbol{\varepsilon}^*) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}^*) = \sigma^2 I.$$

Our model:

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad E(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I.$$

Our $\boldsymbol{\varepsilon}$:

$$\boldsymbol{\varepsilon} = Z\boldsymbol{\gamma} + \boldsymbol{\varepsilon}^*.$$

2.2 Our Results

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{ols} &= (X'X)^{-1}X'\mathbf{Y}, \\ \hat{\mathbf{Y}}_{ols} &= P(X)\mathbf{Y}, \\ RSS &= \mathbf{Y}'(I - P(X))\mathbf{Y}. \end{aligned}$$

Prediction for a future point x_0 :

$$\hat{Y}_{0,ols} = x_0^T \hat{\boldsymbol{\beta}}_{ols}.$$

2.3 Is $\hat{\boldsymbol{\beta}}_{ols}$ Unbiased?

$$E(\hat{\boldsymbol{\beta}}_{ols}) = \boldsymbol{\beta} + \underbrace{(X'X)^{-1}X'Z\boldsymbol{\gamma}}_{\text{bias term}} \quad (\text{biased})$$

Note

If X and Z are orthogonal, i.e., $x'_{(j)}z_{(\ell)} = 0 \quad \forall j = 1, \dots, p+1, \ell = 1, \dots, k$,

$$\text{Var}(\hat{\boldsymbol{\beta}}_{ols}) = \text{Var}((X'X)^{-1}X'\mathbf{Y}) = \sigma^2(X'X)^{-1} \quad [\text{Same as before}]$$

2.4 Bias of the Fitted Values

$$E(\hat{\mathbf{Y}}_{ols}) = P(X)E(\mathbf{Y}) = \underbrace{P(X)X\boldsymbol{\beta}}_{X\boldsymbol{\beta}} + P(X)Z\boldsymbol{\gamma}.$$

Note

If $P(X)Z = Z$, i.e., Z lies in the column space of X , then

$$E(\hat{\mathbf{Y}}_{ols}) = X\boldsymbol{\beta} + Z\boldsymbol{\gamma} = E(\mathbf{Y}).$$

If $P(X)Z = 0$, i.e., Z is orthogonal to $\mathcal{C}(X)$, then

$$E(\hat{\mathbf{Y}}_{ols}) = X\boldsymbol{\beta} \quad \text{is biased.}$$

$$\text{Var}(\hat{\mathbf{Y}}_{ols}) = P(X) \text{Var}(\mathbf{Y}) P(X) = \sigma^2 P(X).$$

2.5 RSS and Estimation of σ^2

Our unbiased estimator of σ^2 is:

$$\frac{RSS}{n - p - 1} = S_x.$$

Is it unbiased?

$$\begin{aligned} E(RSS) &= E(\mathbf{Y}'(I - P(X))\mathbf{Y}) \\ &= E\left[(X\boldsymbol{\beta} + Z\boldsymbol{\gamma} + \boldsymbol{\varepsilon}^*)'(I - P(X))(X\boldsymbol{\beta} + Z\boldsymbol{\gamma} + \boldsymbol{\varepsilon}^*)\right] \\ &= \boldsymbol{\gamma}'Z'(I - P(X))Z\boldsymbol{\gamma} + E[\boldsymbol{\varepsilon}^{*'}(I - P(X))\boldsymbol{\varepsilon}^*] \\ &= \text{tr}((I - P(X))(\sigma^2 I)) + \boldsymbol{\gamma}'Z'(I - P(X))Z\boldsymbol{\gamma} \\ &= \sigma^2(n - p - 1) + \boldsymbol{\gamma}'Z'(I - P(X))Z\boldsymbol{\gamma}. \end{aligned}$$

$$E(S_x) = \sigma^2 + \frac{\boldsymbol{\gamma}'Z'(I - P(X))Z\boldsymbol{\gamma}}{n - p - 1} \geq \sigma^2.$$

Note

S_x estimates the error variance, i.e., variance of $\boldsymbol{\varepsilon}$ ($= Z\boldsymbol{\gamma} + \boldsymbol{\varepsilon}^*$).

The bias in estimating σ^2 will be zero if $(I - P(X))Z = 0$, i.e., Z lies in the column space of X .

If $Z \in \mathcal{C}(X)$, then $Z = XA$, and

$$X\boldsymbol{\beta} + Z\boldsymbol{\gamma} = X\boldsymbol{\beta} + XA\boldsymbol{\gamma} = X(\boldsymbol{\beta} + A\boldsymbol{\gamma}) = X\boldsymbol{\beta}^*.$$

3 Prediction

Suppose (x_0, z_0) is a future observation. We need to estimate Y_0 .

$$Y_0 = x_0'\boldsymbol{\beta} + z_0'\boldsymbol{\gamma} + \varepsilon_0, \quad \varepsilon_0 \sim N(0, \sigma^2).$$

3.1 Prediction Under the Working Model

Our estimate of Y_0 is

$$\hat{Y}_{0,ols} = x_0'\hat{\boldsymbol{\beta}}_{ols} = x_0'(X'X)^{-1}X'\mathbf{Y}.$$

$$E(\hat{Y}_{0,ols}) = x_0'\boldsymbol{\beta} + x_0'(X'X)^{-1}X'Z\boldsymbol{\gamma} \neq x_0'\boldsymbol{\beta} + z_0'\boldsymbol{\gamma}.$$

$$\text{Var}(\hat{Y}_{0,ols}) = \sigma^2 x_0'(X'X)^{-1}X'X(X'X)^{-1}x_0 = \sigma^2 x_0'(X'X)^{-1}x_0. \quad (*)$$

3.2 Prediction Under the Actual (Augmented) Model

Define

$$W = (X : Z), \quad \boldsymbol{\delta} = \begin{bmatrix} \boldsymbol{\beta} \\ \boldsymbol{\gamma} \end{bmatrix}_{(p+k+1)}$$

$$W = n \times (p + k + 1), \quad X = n \times (p + 1), \quad Z = n \times (k)$$

$$\hat{\boldsymbol{\delta}}_{ols} = (W'W)^{-1}W'\mathbf{Y},$$

$$Y_{0,ols}^A = (x'_0 : z'_0) \hat{\boldsymbol{\delta}}_{ols}.$$

$$E(Y_{0,ols}^A) = (x'_0 : z'_0)(W'W)^{-1}W'W\boldsymbol{\delta} = (x'_0 : z'_0)\boldsymbol{\delta} = x'_0\boldsymbol{\beta} + z'_0\boldsymbol{\gamma}.$$

$$\begin{aligned} \text{Var}(Y_{0,ols}^A) &= (x'_0 : z'_0)(W'W)^{-1}W' \text{Var}(\mathbf{Y})W(W'W)^{-1} \begin{pmatrix} x_0 \\ z_0 \end{pmatrix} \\ &= \sigma^2(x'_0 : z'_0)(W'W)^{-1} \begin{pmatrix} x_0 \\ z_0 \end{pmatrix}. \end{aligned} \quad (**)$$

3.3 Block Matrix Inversion

$$W = (X : Z), \quad W'W = \begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z \end{bmatrix}.$$

Formula

$$\begin{bmatrix} A & B \\ B' & D \end{bmatrix}^{-1} \text{ (symmetric)} = \begin{bmatrix} A^{-1} + A^{-1}BH^{-1}B'A^{-1} & -A^{-1}BH^{-1} \\ -H^{-1}B'A^{-1} & H^{-1} \end{bmatrix}$$

where $H = (D - B'A^{-1}B)$, symmetric.

$$(W'W)^{-1} = \begin{bmatrix} \underbrace{(X'X)^{-1} + (X'X)^{-1}X'ZH^{-1}Z'X(X'X)^{-1}}_{(1,1)} & \underbrace{-(X'X)^{-1}X'ZH^{-1}}_{(1,2)} \\ \underbrace{-H^{-1}Z'X(X'X)^{-1}}_{(2,1)} & \underbrace{H^{-1}}_{(2,2)} \end{bmatrix}$$

and

$$H = Z'Z - Z'X(X'X)^{-1}X'Z = Z'(I - P(X))Z = \text{p.s.d. matrix.}$$

3.4 Evaluating (**)

From (**):

$$\begin{aligned} \sigma^2(x'_0 : z'_0)(W'W)^{-1} \begin{pmatrix} x_0 \\ z_0 \end{pmatrix} &= \sigma^2 \left[x'_0(X'X)^{-1}x_0 \right. \\ &\quad + \underbrace{x'_0(X'X)^{-1}X'ZH^{-1}Z'X(X'X)^{-1}x_0}_{(1,1)} \\ &\quad - z'_0H^{-1}Z'X(X'X)^{-1}x_0 \\ &\quad - x'_0(X'X)^{-1}X'ZH^{-1}z_0 \\ &\quad \left. + z'_0H^{-1}z_0 \right] \end{aligned}$$

Let $a_0 = Z'X(X'X)^{-1}x_0$, this simplifies to:

$$= (\alpha) + \sigma^2 \left[a_0' H^{-1} a_0 - 2z_0' H^{-1} a_0 + z_0' H^{-1} z_0 \right].$$